

**DINESH GULATI**  
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<https://hydroagrinexus.com/>

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## EDUCATION

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| <b>Ph.D. in Water Resources</b> , University of Idaho<br><b>GPA:</b> 4.00; <b>Dissertation:</b> <i>Development and deployment of soil water budget models tailored for Idaho agroecosystems and their applications</i><br><b>Advisor:</b> Dr. Meetpal S. Kukal                                                              | 2023-Present |
| <b>Master of Technology in Soil and Water Engineering</b> , Punjab Agricultural University, India<br><b>GPA:</b> 3.32; <b>Thesis:</b> <i>Simulation of groundwater recharge from transplanted and direct seeded rice fields</i><br><b>Advisor:</b> Dr. Sanjay Satpute                                                       | 2018-2021    |
| <b>Bachelor of Technology in Agricultural Engineering</b> , Punjab Agricultural University, India<br><b>GPA:</b> 3.22; <b>Project:</b> <i>Comparative field evaluation of different straw management technologies in combine harvested paddy fields under different straw conditions</i><br><b>Advisor:</b> Dr. M.K. Narang | 2014-2018    |

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## RESEARCH EXPERIENCE

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| <b>Graduate Research Assistant, Department of Soil and Water Systems</b><br><i>University of Idaho, ID</i> <ul style="list-style-type: none"><li>Developed a Google Earth Engine routine to obtain and analyze OpenET, gridMET, and soils data for the Eastern Snake Plain Aquifer.</li><li>Contributed to pyfao56 Open-Source Evapotranspiration and Water Balance Tool (awarded Irrigation Association's Vanguard Award).</li><li>Contributed to developing a Python model for quantifying evapotranspiration of applied water.</li><li>Assessed soil organic carbon gain impacts on agrohydrological processes across soils, climates, and irrigation practices.</li><li>Built SnakeFlux cooperators portal, a Flask-based web app that automates eddy covariance data delivery and visualization to cooperators. (<a href="https://fieldsync.pythonanywhere.com">fieldsync.pythonanywhere.com</a>)</li><li>Built FAO56-Studio, an interactive no-code web-based platform to simulate FAO56-based soil water budgets. (<a href="https://swb.hydroagrinexus.com">swb.hydroagrinexus.com</a>)</li><li>Analysed eddy covariance energy balance closure and flux footprints using the flux-data-qaqc and fluxfootprints packages.</li></ul> | Aug 2024-Present    |
| <b>Summer Research Assistant-IWRRI</b><br><i>University of Idaho, ID</i> <ul style="list-style-type: none"><li>Conducted initial hydrological assessment of Portneuf Watershed for IWRRI.</li><li>Analyzed datasets such as gridMET, PRISM, OpenET, USDA CDL, NLCD, IrrMapper from GEE using JavaScript and Python APIs.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | June-August 2025    |
| <b>Graduate Research Assistant, Department of Agricultural and Biological Engineering</b><br><i>The Pennsylvania State University, PA</i> <ul style="list-style-type: none"><li>Contributed to pyfao56 Open-Source Evapotranspiration and Water Balance Tool</li><li>Developed a machine learning model that enables the usability of NOAA Reference ET forecasts in the Western U.S.</li><li>Assessed soil organic carbon gain impacts on agrohydrological processes across soils, climates, and irrigation practices.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Aug 2023-Aug 2024   |
| <b>Senior Research Fellow, Department of Soil Sciences</b><br>Punjab Agricultural University, India <ul style="list-style-type: none"><li>Planned and managed field experiments under microirrigation systems.</li><li>Developed low-cost, microcontroller-based sensors to observe climate and agricultural parameters for under \$50.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | May 2021 – May 2022 |

- Coordinated with Ph.D. and master's students for research assistance and conducted practical classes for bachelor's degree students.

Aug 2018 –  
Apr 2021

### Master's Research, Department of Soil and Water Engineering

Punjab Agricultural University, India

- Conducted field experiments for direct-seeded and transplanted rice.
- Collected and analyzed soil moisture data using Delta PR2 probes.
- Estimated and compared groundwater recharge using HYDRUS-1D simulations.

## PUBLICATIONS

### Published:

- Keshavarz Afshar, R., Tricarico, J., Medel-Jiménez, F., Kukal, M.S., **Gulati, D.** *et al.* Transparent and harmonized allocation factors for crop-processing co-products in U.S. dairy rations. *Int J Life Cycle Assess* **31**, 180 (2026). <https://doi.org/10.1007/s11367-026-02750-w>
- Gulati, D.**, & Kukal, M. S. (2026). Improved water retention from soil carbon accrual delivers only marginal agrohydrological benefits under most conditions. *Water Resources Research*, 62(4), e2025WR041768. <https://doi.org/10.1029/2025WR041768>
- Thorp, K. R., **Gulati, D.**, Kukal, M., Ames, R., Pokoski, T., & DeJonge, K. C. (2025). Version 1.4.0 - pyfao56: FAO-56 evapotranspiration in Python. *SoftwareX*, 30, 102109. <https://doi.org/10.1016/j.softx.2025.102109>
- Thorp, K. R., DeJonge, K. C., Pokoski, T., **Gulati, D.**, Kukal, M., Farag, F., ... & Holzkaemper, A. (2024). Version 1.3. 0 pyfao56: FAO-56 evapotranspiration in Python. *SoftwareX*, 26, 101724. <https://doi.org/10.1016/j.softx.2024.101724>
- Singla, P., Sharda, R., Sharma, S., **Gulati, D.**, Pandey, K., Navpreem, S., ... & Sharma, A. (2023). Variation in physio-chemical attributes and WUE during growth and development of Pak choi (*Brassica rapa* L. subsp. *chinensis* L.) under different drip fertigation and mulching treatments. *Acta Alimentaria*, 52(3), 458-468. <https://doi.org/10.1556/066.2023.00084>
- Gulati, D.**, Satpute, S., Kaur, S., & Aggarwal, R. (2022). Estimation of potential recharge through direct seeded and transplanted rice fields in semi-arid regions of Punjab using HYDRUS-1D. *Paddy and Water Environment*, 1-14. <https://doi.org/10.1007/s10333-021-00876-1>

## PRESENTATIONS

- Assimilation of in-situ soil moisture to constrain FAO-56 crop evapotranspiration in water-limited environments. July 14, 2026, Indianapolis, IN. **(ASABE)**
- Quantifying the water footprints of U.S. dairy feed across the CONUS. July 14, 2026, Indianapolis, IN. **(ASABE)** [Poster]
- FAO56-Studio: An interactive, no-code online platform for simulating FAO-56 based soil water budgets. July 15, 2026, Indianapolis, IN. **(ASABE)**
- Evaluating feasibility of deficit irrigation in Idaho's Magic Valley. March 3, 2026, Boise, ID **(IDEQ IWQW)** [Poster]
- Revisiting potato evapotranspiration: Findings from continuous flux measurements in a commercial production field. Jan 21, 2026, Pocatello, ID. **(Idaho Potato Conference)**
- Increased water retention from soil carbon gain is insufficient to result in a tangible positive change in agricultural water budgets. Nov 11, 2025, SLC, UT. **(CANVAS)**
- Assimilation of in-situ soil moisture to constrain FAO-56 crop evapotranspiration in water-limited environments. Nov 11, 2025, SLC, UT. **(CANVAS)**
- Consumptive use & crop yield response to irrigation demand management in Idaho's Magic Valley. Nov 4, 2025, Boise, ID. **(PNW Summit)** [Poster]
- Increased water retention from soil carbon gain is insufficient to result in a tangible positive change in agricultural water budgets. July 15, 2025, Toronto, Ontario, Canada. **(ASABE/CSABE)**
- Deploying OpenET to access consumptive use from field to water entity scales across Eastern Snake Plain Aquifer. July 15, 2025, Toronto, Ontario, Canada. **(ASABE/CSABE)**
- Tracking evapotranspiration of applied water and effective precipitation by color-coding standardized soil water budgets. July 14, 2025, Toronto, Ontario, Canada. **(ASABE/CSABE)**
- How NRCS conservation practice standards impact agricultural water budgets? March 3, 2025, Boise, ID **(IDEQ IWQW)** [Poster]
- Hydrological losses under increased soil organic carbon and weather variability in different soil textures, Nov 11, 2024, San Antonio, TX **(ASA, CSSA, SSSA)**

14. How does improving soil structure impact crop evapotranspiration and water stress across soil textures and aridity? Nov 11, 2024, San Antonio, TX (**ASA, CSSA, SSSA**)
  15. Climatic and Seasonal Variation in  $ET_r$  to  $ET_o$  ratios calculated using ASCE standardized Penman-Monteith across the contiguous U.S., July 29, 2024, Anaheim, CA (**ASABE**) [Poster]
  16. Translating soil carbon sequestration into agrohydrological outcomes across a spectrum of aridity and soil texture, July 30, 2024, Anaheim, CA (**ASABE**)
  17. Climatic and Seasonal Variation in  $ET_r$  to  $ET_o$  ratios calculated using ASCE standardized Penman-Monteith across the contiguous U.S., July 15, 2024, State College, PA (**NABEC**) [Poster]
  18. Translating soil carbon sequestration into agrohydrological outcomes across a spectrum of aridity and soil texture, July 16, 2024, State College, PA (**NABEC**)
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## RESEARCH PROPOSALS

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1. Beyond flux towers: evaluating ET models for irrigation decisions using soil water depletion as a validation target. 2026, Meter Group (\$10,000 worth of instruments)
  2. Understanding consumptive use on the ESPA: Data gaps, uncertainty, and impacts, PI: Meetpal Kukal, 2025 Idaho Water Research Priorities, IWRRI (\$44,870); Role: Key personnel.
  3. Hydrologic modelling and data provisioning for development of a dairy water footprint tool for the Western U.S., PI: Meetpal Kukal, Dairy Management Inc. (\$86,875); Role: Key personnel.
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## HONORS AND AWARDS

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- Grant A. Harris Fellowship (\$10,000 worth of instruments) in 2026 from Meter Group
  - Vanguard Award to pyfao56 in 2025 from Irrigation Association
  - Joseph Jordan Student Research Fellowship (\$3500) in 2025 from IWRRI
  - Whiting Water Resources Scholarship (\$1,000) in 2024 from the University of Idaho
  - Irrigation E3 learner in 2024 from Irrigation Association
  - Dr. S.D. Khepar Gold Medal in 2021 from Punjab Agricultural University
  - Best M.Tech Thesis in Agricultural Engineering in 2022 from ISTE National Award
  - Merit scholarships during Bachelor's and Master's studies from Punjab Agricultural University
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## PROFESSIONAL DEVELOPMENT

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- Completed a three-day APSIM workshop on crop management simulation at Iowa State University
- Specialization in Deep Learning by deeplearning.ai on Coursera
- Machine Learning with Python & Fundamentals of R programming on Udemy
- Attended IoT workshop on precision farming by NCPAH, Ministry of Agriculture, India